

Ikhan Choi

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EDUCATION

- **The University of Tokyo** Apr 2025 – Now
Ph.D. program in Mathematical Science
- **The University of Tokyo** Apr 2023 – Mar 2025
M.S. in Mathematical Science
- **Pohang University of Science and Technology** Mar 2016 – Aug 2022
B.S. in Mathematics
(Early graduation: leave of absence for three years including military service)
- **Seoul Science High School** Mar 2013 – Feb 2016
(Specialized high school for gifted students)

RESEARCH INTERESTS

Functional analytic features of bivariant K-theories and noncommutative topology from the viewpoint of homotopy theory, and their applications to representation theory and mathematical physics.

PUBLICATIONS

- **I. Choi**, *A solution to Haagerup's problem and positionve Hahn-Banach separation theorems in operator algebras*, J. Funct. Anal. 290 (2026), no. 2, Paper No. 111228, 11 pp. Master's thesis
- **I. Choi**, *Curved folding and planar cutting of simple closed curve on a conical origami*, Kodai Math. J. 39 (2016), no. 3, 579–595. High school thesis

AWARDS AND HONORS

- Research Fellowship for Young Scientists (Doctoral Course) Apr 2026 – Mar 2028
Japan Society for the Promotion of Science (JSPS)
- Dean's Award Mar 2025
M.S. in Graduate School of Mathematical Science at the University of Tokyo
- Japanese Government (MEXT) Scholarship Apr 2023 – Mar 2026
Embassy recommendation, Research student
- Gold Prize in 38th Mathematics Competition for University Students in Korea Nov 2019
1st group for math majors
Sponsored by Korean Mathematics Society

TALKS

- *KK-theory as a stable infinity category* 3 Sep 2025
Matsuyama, Functional Anaysis Conference (Junior)
- *A solution to Haagerup's problem on normal weights*
Kyoto, Operator Algebraists' Symposium: Recent Developments in Operator Algebras (invited) 24 Sep 2025
Seoul, KIAS Seminars (invited) 14 Jul 2025
Seoul, China-Japan-Korea Conference on Functional Analysis 8 July 2025
Kyoto, Kyoto Operator Algebra Seminars (invited) 6 Jun 2025
Tokyo, Operator Algebra Seminars at the University of Tokyo (invited) 13 May 2025
- *Unbounded KK-theory* 18 Sep 2025
Sapporo, Functional Anaysis Conference (Junior)

TEACHING

- Teaching Assistant: Linear Algebra I Spring 2026
- Teaching Assistant: Linear Algebra II Fall 2025
- Teaching Assistant: Geometry II Fall 2024
- Teaching Assistant: Complex Analysis I Fall 2023
- Student Mentoring: MATH 423 Introduction to Differential Geometry Fall 2019

OTHER SKILLS AND QUALIFICATIONS

- Korean (native), Japanese (fluent), English (proficient)